HEALTHIER, LONGER, BETTER LIVES

Medical Receipt OCR Benchmark

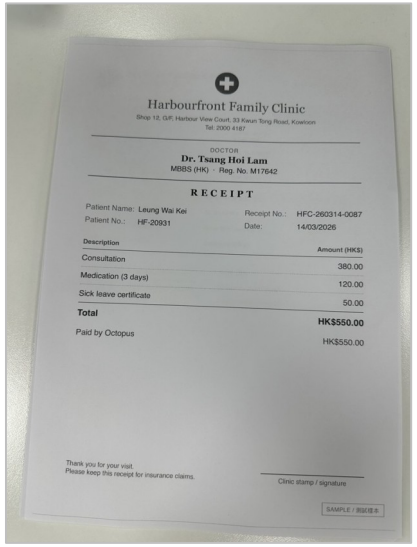
Can open-source OCR with rules match Azure AI at extracting receipt fields?

9 receipts · 6 fields · 4 approaches

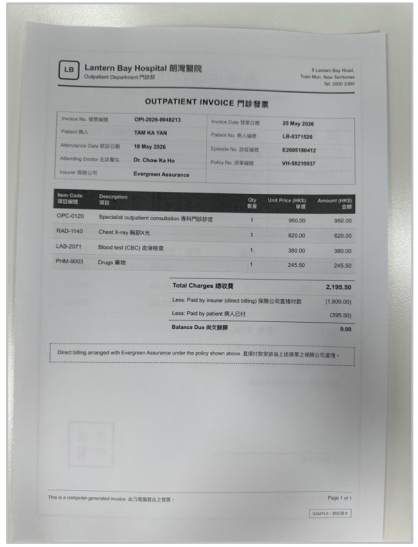
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What We Tested

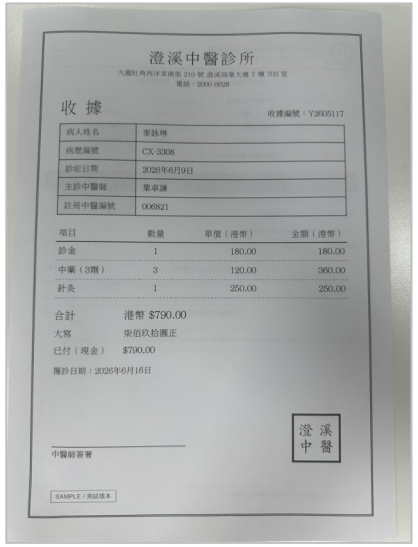
9 receipts with different layouts and languages. Each approach reads the text, then finds 6 fields.



English



Bilingual



Chinese

Approach	1. Read text	2. Find fields
PaddleOCR Light	Open-source OCR	Rules
PaddleOCR Heavy	Open-source OCR	Rules
Azure DI	Azure OCR	Rules
Azure DI + AOAI	Azure OCR	Azure OpenAI

Fields: doctor, registration no., patient, receipt no., service date, total.
51 fields scored.

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Results: Field Accuracy and Speed

Improving the rules raised PaddleOCR Light from 37.3% to 88.2%. AOAI reached 88.2% without changes.

Approach	Original rules	Improved rules*	Time per receipt
PaddleOCR Light	37.3% (19/51)	**88.2%** (45/51)	13.9 s
PaddleOCR Heavy	29.4% (15/51)	64.7% (33/51)	53.3 s
Azure DI	41.2% (21/51)	**86.3%** (44/51)	3.8 s
Azure DI + AOAI	**88.2%** (45/51)	Not applicable (prompt unchanged)	6.2 s

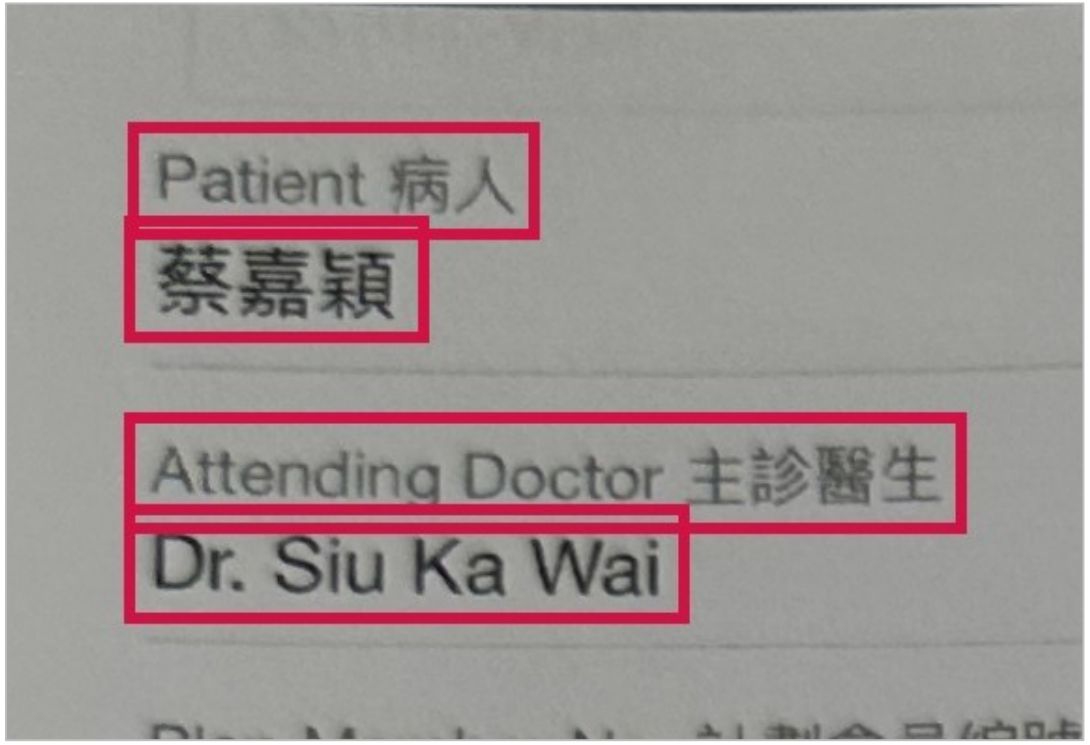
** Tuned using these 9 receipts, so not a measure of accuracy on new receipts. Accuracy = correct fields out of 51. All approaches processed all 9 receipts.*

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Why Accuracy Improved: The Rules, Not the OCR

The OCR read the text correctly. The original rules picked the wrong text.



Field	Correct	Original rules	Improved rules	DI + AOAI
Patient	蔡嘉穎	✗ 病人	✓ 蔡嘉穎	✓ 蔡嘉穎
Doctor	Dr. Siu Ka Wai	✗ not found	✓ Dr. Siu Ka Wai	✓ Dr. Siu Ka Wai

- Original rules: took the Chinese label 病人 ("patient") as the patient's name.
- Improved rules: added bilingual labels, so the label and the value are kept apart.

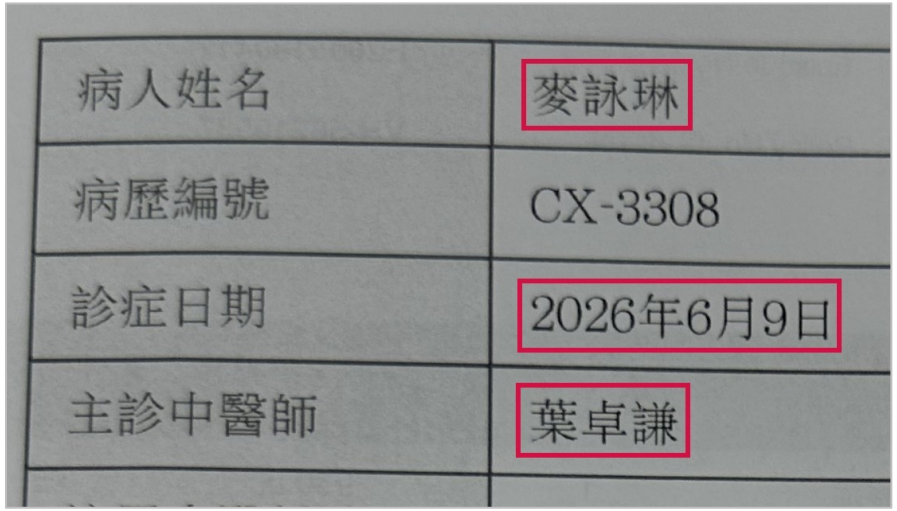
Bilingual medical centre receipt. Rules shown on PaddleOCR Light text; Azure DI with the original rules made the same patient error.

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The Trade-off: Rules Need Manual Maintenance

High accuracy came from rewriting the rules by hand, and one fix broke another receipt.



Chinese medicine receipt: original rules 0/6 fields, improved rules 6/6.

69 → 225

label phrases in the rules, added by hand after reviewing the errors

369303 → 380048

vet invoice receipt number: correct before the changes, wrong after (a regression)

New layouts

may need more rule changes and retesting of every receipt

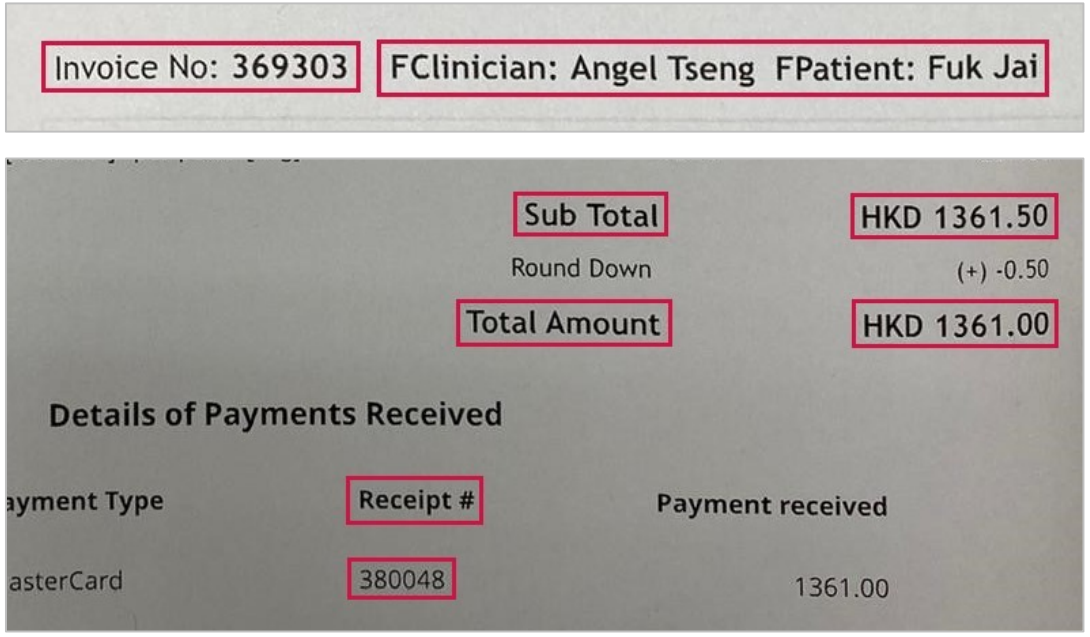
AOAI used the same prompt for every receipt. Whether that means less maintenance on new layouts still needs to be tested.

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Where Errors Remain: Both Approaches Make Mistakes

On the vet invoice, the improved rules and AOAI each got a different field wrong.



Field	Correct	Original rules	Improved rules	DI + AOAI
Receipt no.	369303	✓ 369303	✗ 380048	✓ 369303
Patient	Fuk Jai	✗ not found	✗ not found	✓ Fuk Jai
Total	1361	✓ 1361.0	✓ 1361.0	✗ 1361.5

- Improved rules: picked the payment receipt number.
- Rules: the OCR read "FPatient:", so the patient label was missed.
- AOAI: picked the sub total instead of the total.

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Practical Comparison of the Two Candidates

Similar accuracy on these receipts; they differ in cost, speed and how new layouts are handled.

	PaddleOCR Light + improved rules	Azure DI + AOAI
Accuracy (these 9 receipts)	88.2% (45/51), tuned	88.2% (45/51), prompt unchanged
Time per receipt	13.9 s	6.2 s
Where it runs	Our own compute, offline	Azure cloud services
Cost per receipt	CPU time only, no API fee	DI page + about 602 input / 263 output tokens (prices to confirm)
New receipt layouts	Edit rules and retest	Same prompt; still to be tested
Main risk	Maintenance and regressions	API cost and model variability

Both approaches read the receipt text well; the differences are in finding the fields.

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Conclusion and Next Steps

Rules can reach high accuracy, but only with ongoing manual work. That is the decision to make.

The question is not only which method is most accurate.

It is whether we want to keep updating extraction rules each time an unfamiliar receipt layout appears, or pay per request for a model that may handle new layouts without changes.

1. Freeze both

Lock the improved rules and the AOAI prompt.

2. Test on new receipts

30+ real, anonymised receipts not seen before. Count the rule changes needed.

3. Confirm prices

Compute, Document Intelligence pages and gpt-5-mini tokens.

Limitation: 9 receipts, mostly printed mock receipts; the improved rules were tuned on them. Candidates for further testing: PaddleOCR Light + improved rules, and Azure DI + AOAI.

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